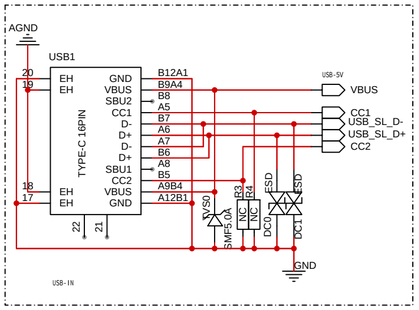
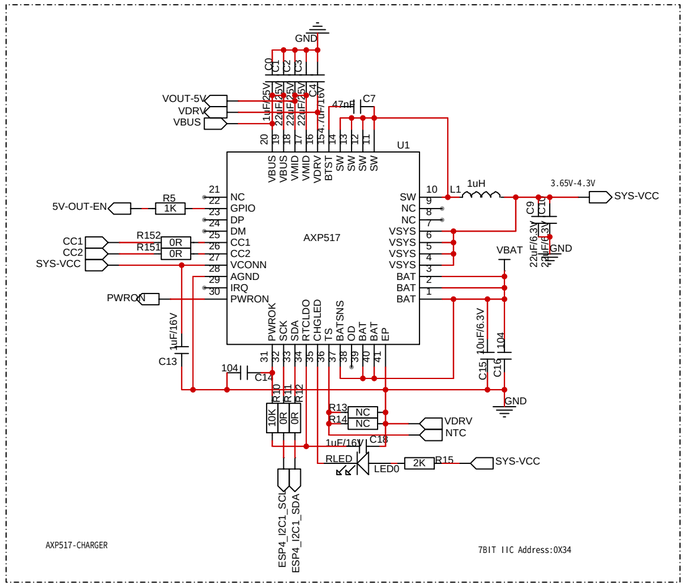
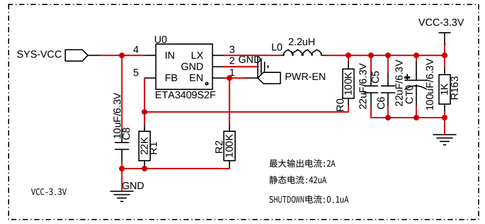
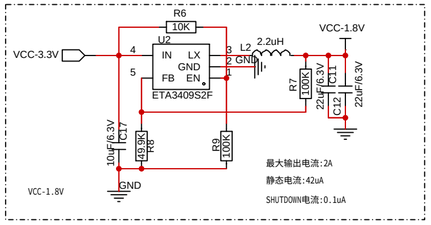
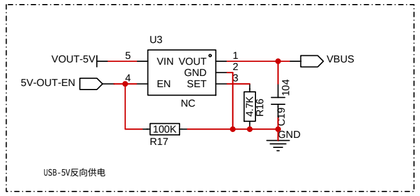
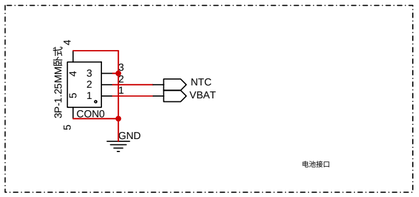

AGND
USB1
TYPE-C 16PIN
EH
EH
GND
VBUS
SBU2
CC1
D-
D+
D-
D+
SBU1
CC2
VBUS
GND
B12A1
B9A4
B8
A5
B7
A6
A7
B6
A8
B5
A9B4
A12B1
USB-5V
VBUS
CC1
USB_SL_D-
USB_SL_D+
CC2
R3
R4
NC
NC
ESD
DC0
DC1
SMF5.0A
5V50
GND
USB-IN

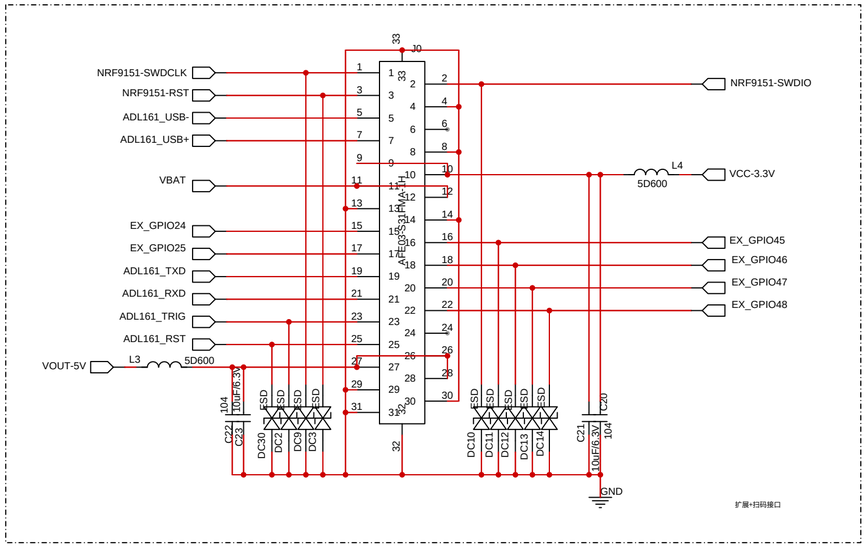

GND
VOUT-5V
VDRV
VBUS
C1
C2
C3
10uF/25V
22uF/6.3V
22uF/6.3V
22uF/6.3V
C4
4.7uF/16V
20
19
18
17
16
15
C7
47nF
U1
VBUS
VBUS
VMID
VMID
VDRV
BTST
SW
SW
SW
SW
14
13
12
11
5V-OUT-EN
R5
1K
21
22
23
24
25
26
27
28
29
30
NC
GPIO
DP
DM
CC1
CC2
VCONN
AGND
IRQ
PWRON
AXP517
CC1
CC2
SYS-VCC
R152
R151
0R
0R
PWRON
1uF/16V
C13
31
32
33
34
35
36
37
38
39
40
41
PWROK
SCK
SDA
RTCLDO
CHGLED
TS
BATSNS
OUT
BAT
BAT
EP
104
C14
R10
10K
R11
10K
R12
10K
SW
NC
NC
BAT
BAT
BAT
10
9
8
7
6
5
4
3
2
1
L1
1uH
3.65V-4.3V
SYS-VCC
C9
C11
22uF/6.3V
22uF/6.3V
VBAT
GND
C15
10uF/6.3V
104
C15
GND
R13
R14
NC
NC
VDRV
NTC
C18
1uF/16V
RLED
LED0
2K
R15
SYS-VCC
ESP4_I2C1_SCL
ESP4_I2C1_SDA
AXP517-CHARGER
7BIT IIC Address:0X34

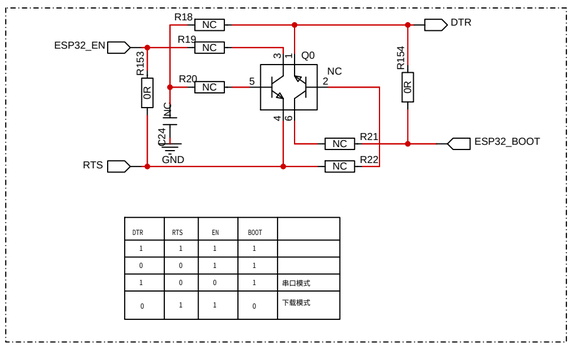

SYS-VCC
U0
IN
LX
GND
FB
EN
ETA3409S2F
4
5
3
2 GND
1
L0
2.2uH
PWR-EN
VCC-3.3V
R4
100K
22uF/6.3V
C5
22uF/6.3V
C10
100uF/6.3V
1K
R103
10uF/6.3V
C3
R1
22K
R2
100K
GND
VCC-3.3V
最大输出电流:2A
静态电流:42uA
SHUTDOWN电流:0.1uA

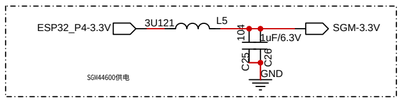

R6
10K
VCC-3.3V
U2
IN
LX
GND
FB
EN
ETA3409S2F
4
5
3
2 GND
1
L2
2.2uH
VCC-1.8V
R7
100K
22uF/6.3V
C11
C12
22uF/6.3V
10uF/6.3V
C17
49.9K
R8
R9
100K
GND
VCC-1.8V
最大输出电流:2A
静态电流:42uA
SHUTDOWN电流:0.1uA

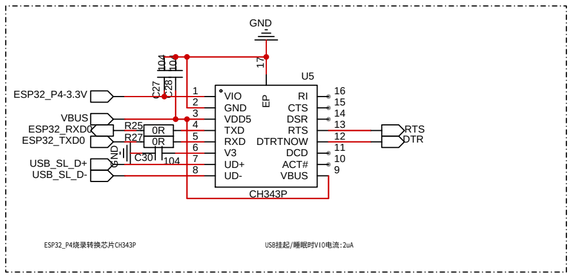

U3
VOUT-5V
5V-OUT-EN
VIN
EN
VOUT
GND
SET
NC
5
4
1
2
3
VBUS
100K
R17
4.7K
R16
104
C19
GND

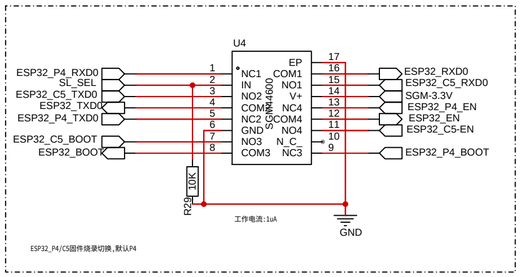

3P-1.25MMB式
4
3
2
1
5
CON0
NTC
VBAT
GND
电池接口

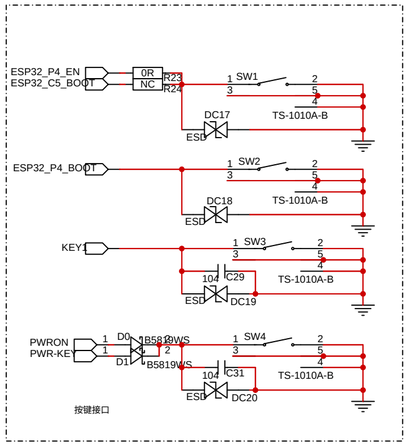

J0
33
NRF9151-SWDCLK
NRF9151-RST
ADL161_USB-
ADL161_USB+
VBAT
EX_GPIO24
EX_GPIO25
ADL161_TXD
ADL161_RXD
ADL161_TRIG
ADL161_RST
VOUT-5V
L3
5D600
NRF9151-SWDIO
L4
5D600
VCC-3.3V
EX_GPIO45
EX_GPIO46
EX_GPIO47
EX_GPIO48
104
10uF/6.3V
C22
C23
DC30
DC2
DC9
DC3
DC10
DC11
DC12
DC13
DC14
C21
10uF/6.3V
104
C20
GND
扩展+扫码接口

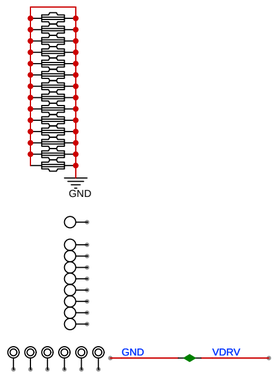

R18
NC
DTR
R19
NC
ESP32_EN
R153
0R
R20
NC
Q0
NC
R154
0R
NC
C24
GND
NC
R21
ESP32_BOOT
RTS
NC
R22
DTR
RTS
EN
BOOT
1
1
1
1
0
0
1
1
1
0
1
0
0
1
串口模式
0
1
1
0
下载模式


ESP32_P4-3.3V
3U121
L5
SGM-3.3V
104
C21
1uF/6.3V
C28
GND
SGM44600供电

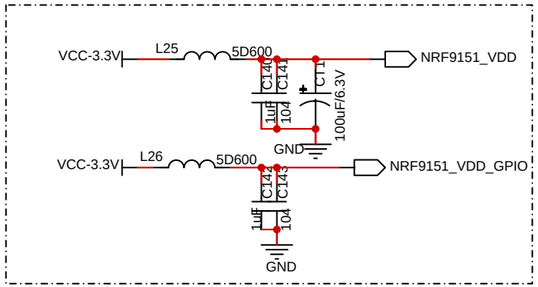

GND
104
C27
104
C28
17
U5
ESP32_P4-3.3V
VBUS
ESP32_RXD0
ESP32_TXD0
USB_SL_D+
USB_SL_D-
R25
0R
R27
0R
C30
104
VIO
GND
VDD5
TXD
RXD
V3
UD+
UD-
EP
RI
CTS
DSR
RTS
DTR
NOW
DCD
ACT#
VBUS
RTS
DTR
CH343P
ESP32_P4烧录转换芯片CH343P
USB挂起/睡眠时VIO电流:2uA

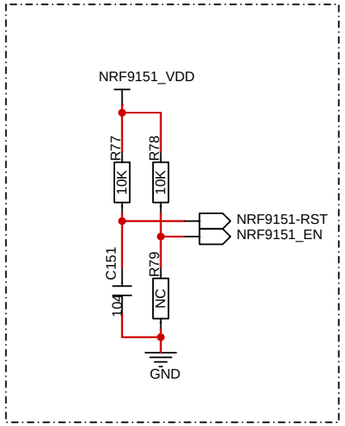

U4
ESP32_P4_RXD0
SL_SEL
ESP32_C5_TXD0
ESP32_TXD0
ESP32_P4_TXD0
ESP32_C5_BOOT
ESP32_BOOT
NC1
IN
NO2
COM2
NC2
GND
NO3
COM3
EP
COM1
NO1
V+
NC4
COM4
NO4
N_C
NC3
SGM44600
ESP32_RXD0
ESP32_C5_RXD0
SGM-3.3V
ESP32_P4_EN
ESP32_EN
ESP32_C5-EN
ESP32_P4_BOOT
R29
10K
工作电流:1uA
GND
ESP32_P4/C5固件烧录切换,默认P4

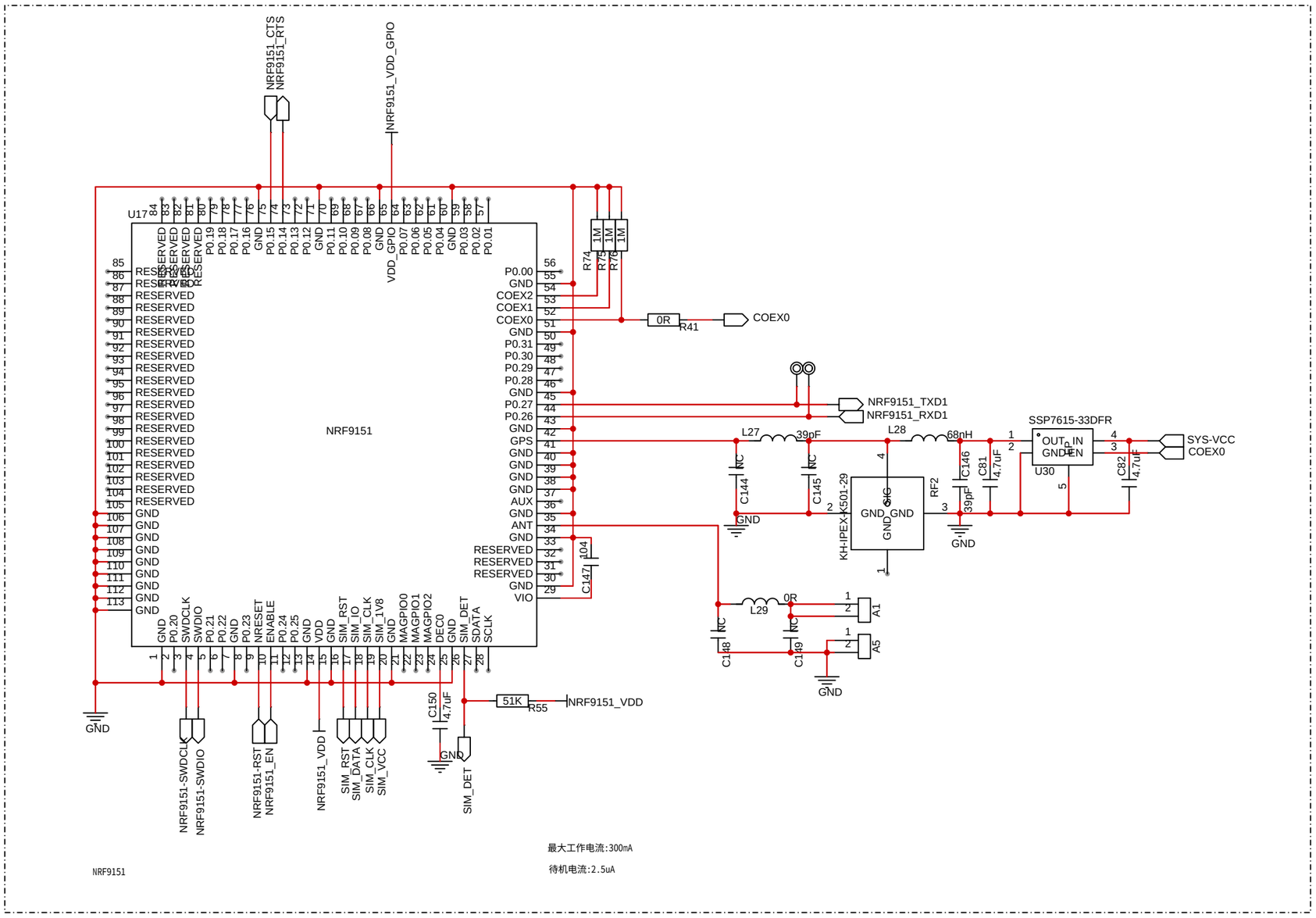

ESP32_P4_EN
ESP32_C5_BOOT
0R
NC
R23
R24
SW1
TS-1010A-B
DC17
ESD
ESP32_P4_BOOT
SW2
TS-1010A-B
DC18
ESD
KEY1
SW3
104
C29
TS-1010A-B
DC19
ESD
PWRON
PWR-KEY
D0
B5819WS
D1
B5819WS
SW4
104
C31
TS-1010A-B
DC20
ESD
按键接口

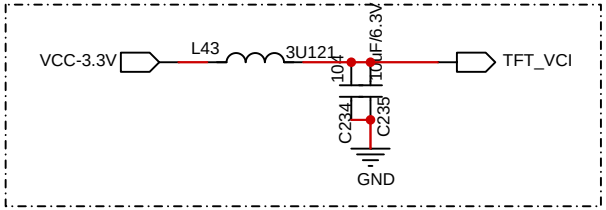

GND
GND
VDRV

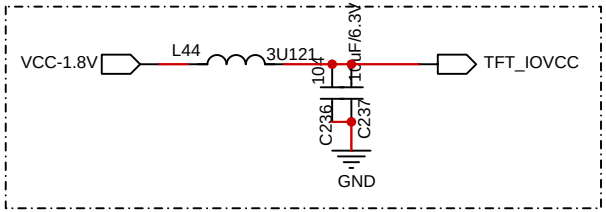
ESP32-P4供电
最大输出电流:800mA
静态电流:3uA
40Mhz晶振
32.768Khz晶振
P4状态指示灯
ESP32_P4_VDD_HP
I2C Address:0x20
GP10扩展
待机电流:0.25uA
ESP32-P4X
最大工作电流:600mA
Lght-sleep工作电流:20uA--3.5mA
Deep-sleep工作电流:5uA--25uA
CPU-EN脚置低工作电流:2uA
ESP32-P4-FLASH/128M
USB-PHY供电
SD卡供电
nRF9151-SIM/ESP32_P4-CARD

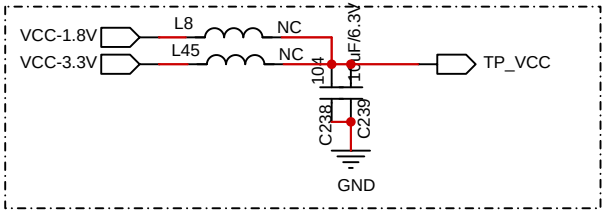

VCC-3.3V
L25
5D600
C140
1uF
C141
104
C142
100uF/6.3V
NRF9151_VDD
GND
VCC-3.3V
L26
5D600
C142
1uF
C143
104
NRF9151_VDD_GPIO
GND

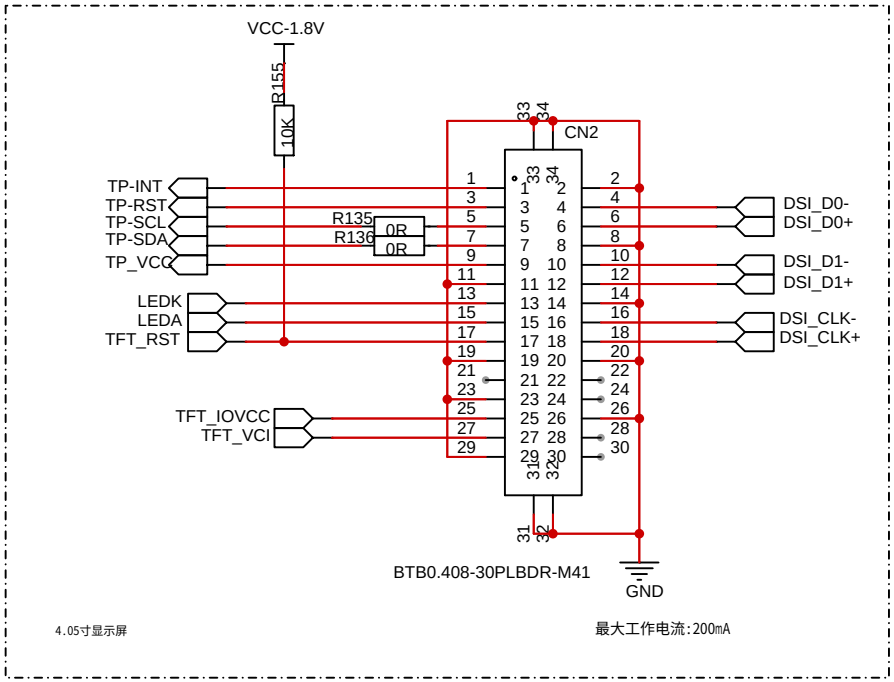

NRF9151_VDD
R77
10K
R78
10K
NRF9151-RST
NRF9151_EN
C151
104
R79
NC
GND

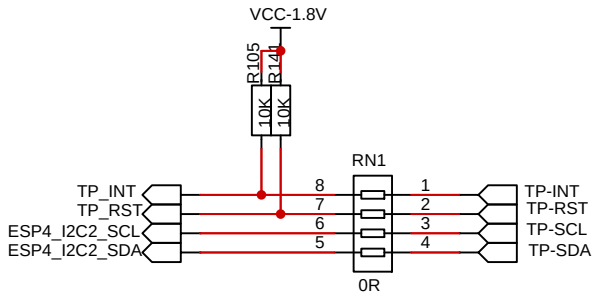

NRF9151_CTS
NRF9151_RTS
NRF9151_VDD_GPIO
U17
NRF9151
R74
1M
R75
1M
R76
1M
0R
R41
COEX0
NRF9151_TXD1
NRF9151_RXD1
L27
39pF
L28
68nH
SSP7615-33DFR
OUT
GND
IN
EN
U30
SYS-VCC
COEX0
C144
NC
C145
NC
C146
4.7uF
C81
4.7uF
C82
4.7uF
KH-IPEX-K501-29
RF2
39pF
C147
104
L29
0R
A1
A5
C148
NC
C149
NC
C150
4.7uF
51K
R55
NRF9151_VDD
SIM_DET
NRF9151-SWDCLK
NRF9151-SWDIO
NRF9151-RST
NRF9151_EN
NRF9151_VDD
SIM_RST
SIM_DATA
SIM_CLK
SIM_VCC
最大工作电流:300mA
待机电流:2.5uA
NRF9151

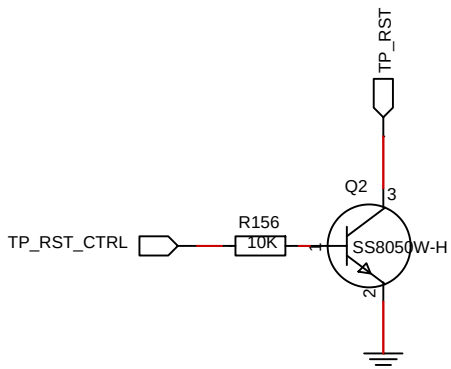

VCC-3.3V
L43
3U121
104
10uF/6.3V
TFT_VCI
C234
C235
GND

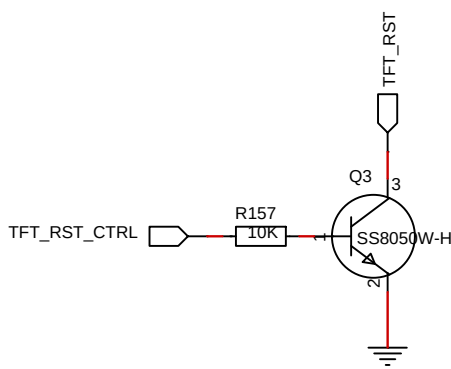

VCC-1.8V
L44
3U121
104
10uF/6.3V
TFT_IOVCC
C236
C237
GND

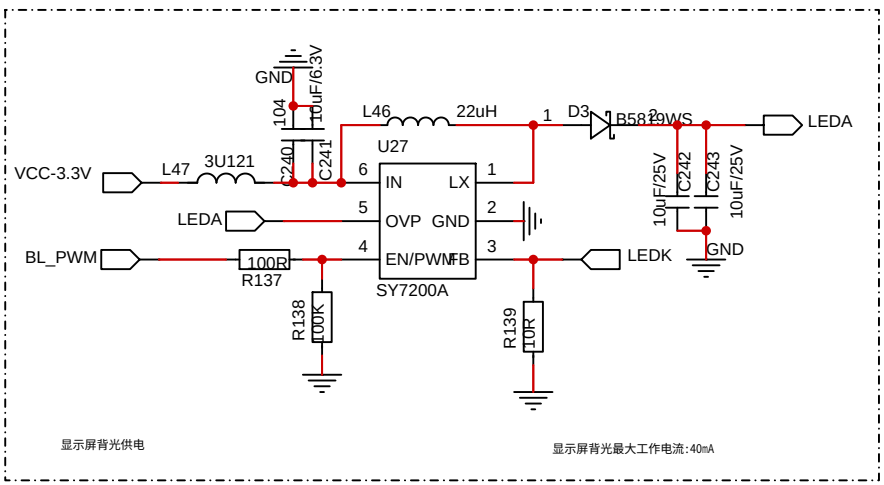

VCC-1.8V
VCC-3.3V
L8
L45
NC
NC
104
10uF/6.3V
TP_VCC
C238
C239
GND

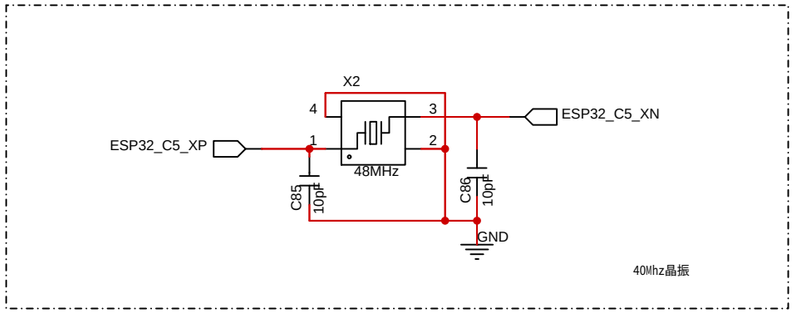

VCC-1.8V
R155
10K
CN2
TP-INT
TP-RST
TP-SCL
TP-SDA
TP_VCC
R135
R136
0R
0R
LEDK
LEDA
TFT_RST
TFT_IOVCC
TFT_VCI
DSI_D0-
DSI_D0+
DSI_D1-
DSI_D1+
DSI_CLK-
DSI_CLK+
BTB0.408-30PLBDR-M41
GND
4.05寸显示屏
最大工作电流:200mA

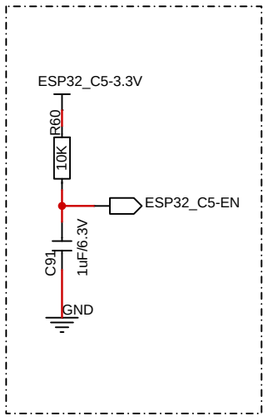

VCC-1.8V
R105
R141
10K
10K
RN1
TP_INT
TP_RST
ESP4_I2C2_SCL
ESP4_I2C2_SDA
0R
TP-INT
TP-RST
TP-SCL
TP-SDA

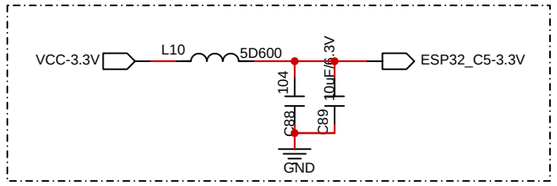

TP_RST
Q2
TP_RST_CTRL
R156
10K
SS8050W-H

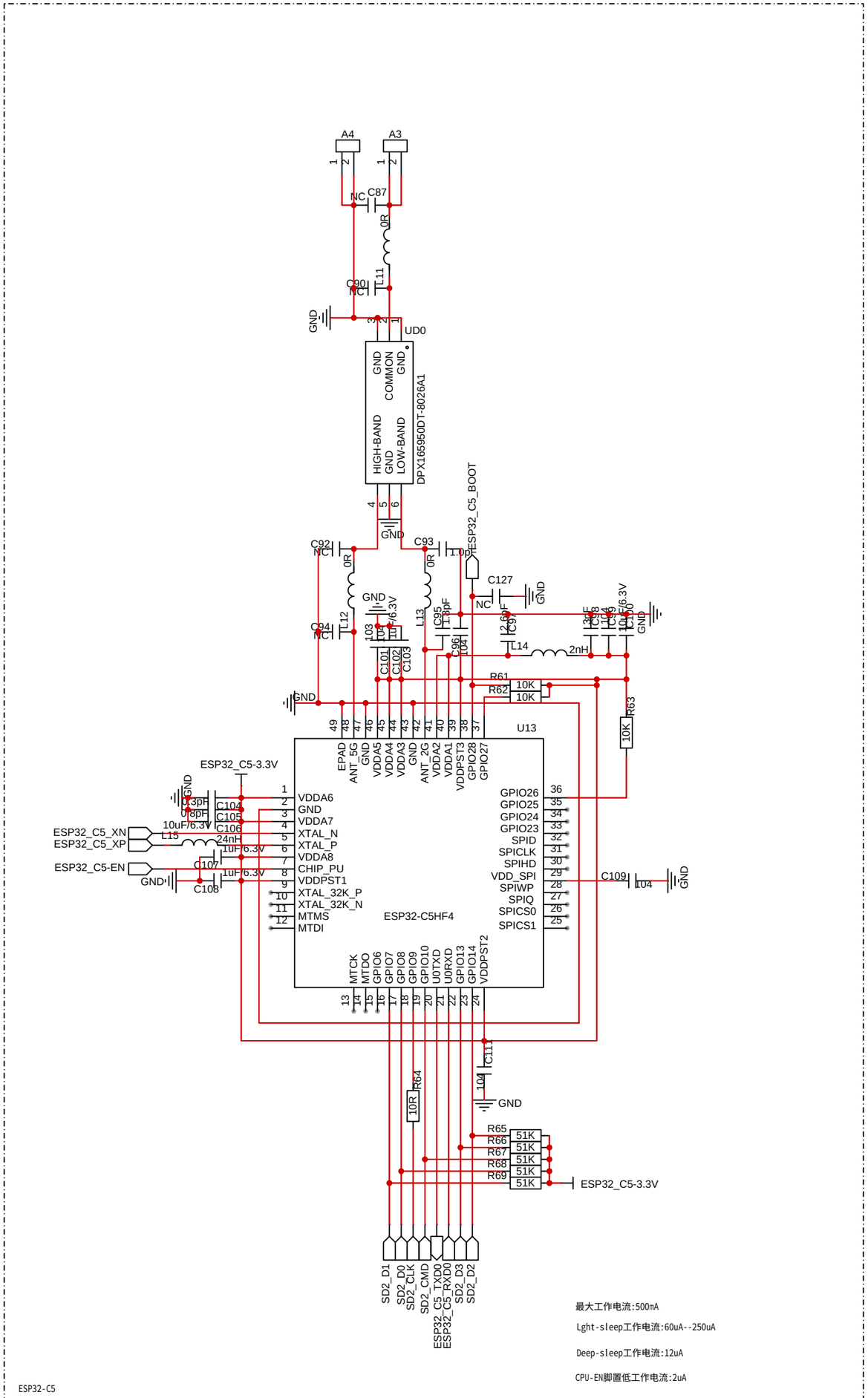

TFT_RST
Q3
TFT_RST_CTRL
R157
10K
SS8050W-H

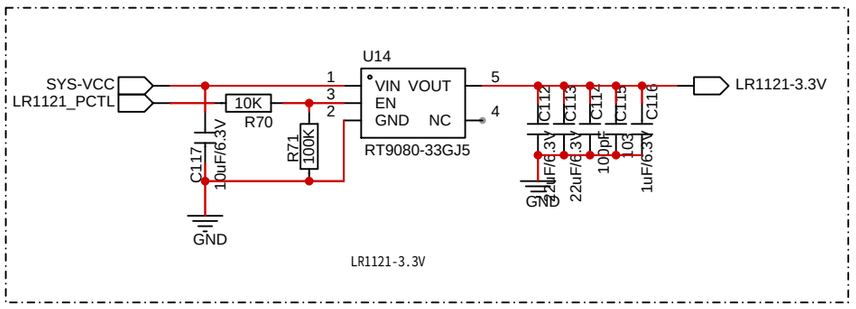

GND
104
10uF/6.3V
L46
22uH
D3
B5819WS
LEDA
U27
VCC-3.3V
L47
3U121
C240
C241
IN
LX
OVP
GND
EN/PWM
FB
LEDA
BL_PWM
100R
R137
SY7200A
LEDK
10uF/25V
C242
C243
10uF/25V
GND
R138
100K
R139
10R
显示屏背光供电
显示屏背光最大工作电流:40mA

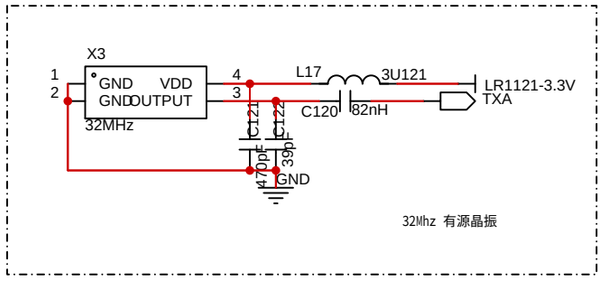

X2
4
3
1
2
ESP32_C5_XP
ESP32_C5_XN
48MHz
C85
10pF
C86
10pF
GND
40Mhz晶振

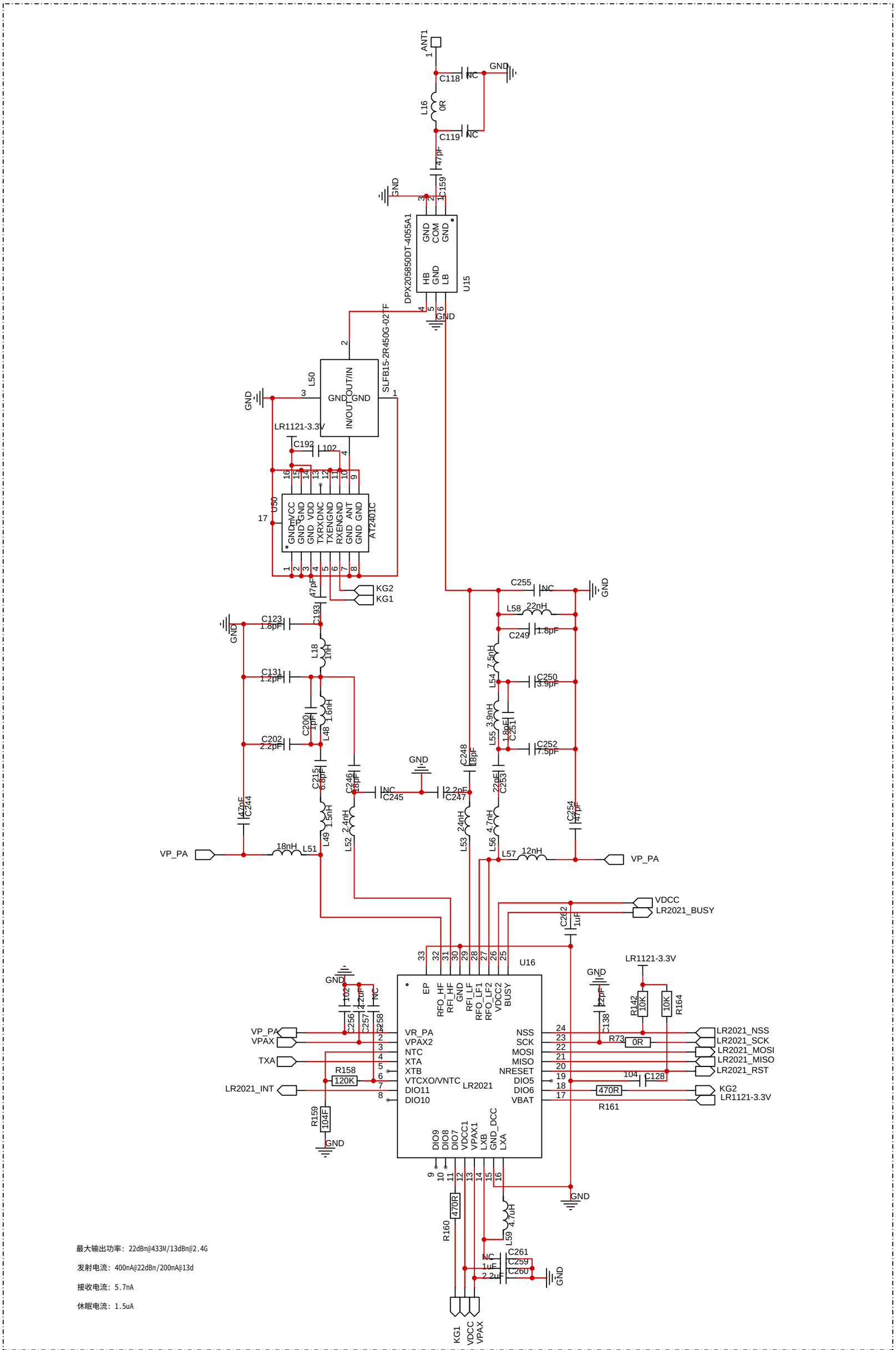

ESP32_C5-3.3V
R60
10K
ESP32_C5-EN
C91
1uF/6.3V
GND


VCC-3.3V
L10
5D600
ESP32_C5-3.3V
104
C88
10uF/6.3V
C89
GND


A4
A3
NC
C87
0R
C90
NC
L11
GND
UD0
GND
COMMON
GND
HIGH-BAND
GND
LOW-BAND
DPX165950DT-8026A1
ESP32_C5_BOOT
C92
NC
0R
C93
0R
1.0nF
C127
NC
GND
C94
NC
L12
GND
103
C95
1.8pF
L13
C96
104
C97
12.6pF
C98
3nF
C99
104
C100
10uF/6.3V
GND
C101
C102
C103
L14
2nH
R61
R62
10K
10K
GND
R63
10K
U13
EPAD
ANT_5G
GND
VDDA5
VDDA4
VDDA3
GND
ANT_2G
VDDA2
VDDA1
VDDPST3
GPIO28
GPIO27
ESP32_C5-3.3V
GND
0.3pF
C104
0.8pF
C105
10uF/6.3V
C106
ESP32_C5_XN
ESP32_C5_XP
L15
24nH
10uF/6.3V
C107
ESP32_C5-EN
GND
10uF/6.3V
C108
VDDA6
GND
VDDA7
XTAL_N
XTAL_P
VDDA8
CHIP_PU
VDDPST1
XTAL_32K_P
XTAL_32K_N
MTMS
MTDI
ESP32-C5HF4
GPIO26
GPIO25
GPIO24
GPIO23
SPID
SPICLK
SPIHD
VDD_SPI
SPIWP
SPIQ
SPICS0
SPICS1
C109
104
GND
MTCK
MTDO
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
U0TXD
U0RXD
GPIO13
GPIO14
VDDPST2
C111
104
GND
R64
10R
R65
R66
R67
R68
R69
51K
51K
51K
51K
51K
ESP32_C5-3.3V
SD2_D1
SD2_D0
SD2_CLK
SD2_CMD
ESP32_C5_TXD0
ESP32_C5_RXD0
SD2_D3
SD2_D2
最大工作电流:500mA
Lght-sleep工作电流:60uA--250uA
Deep-sleep工作电流:12uA
CPU-EN脚置低工作电流:2uA
ESP32-C5



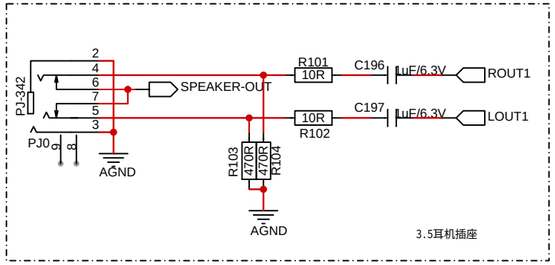



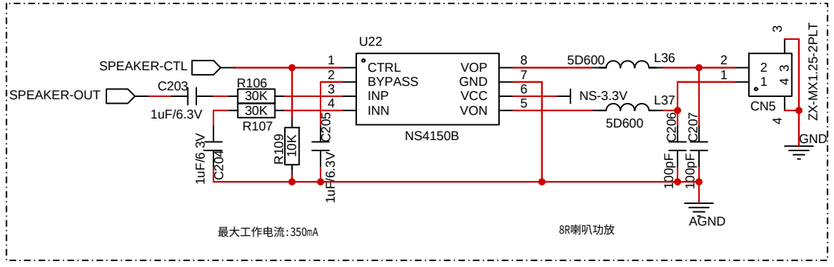

最大输出功率：22dBm@433M/13dBm@2.4G

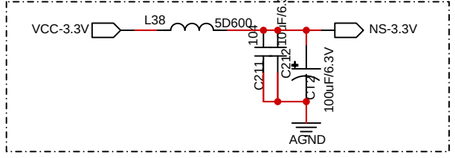
发射电流：400mA@22dBm/200mA@13d

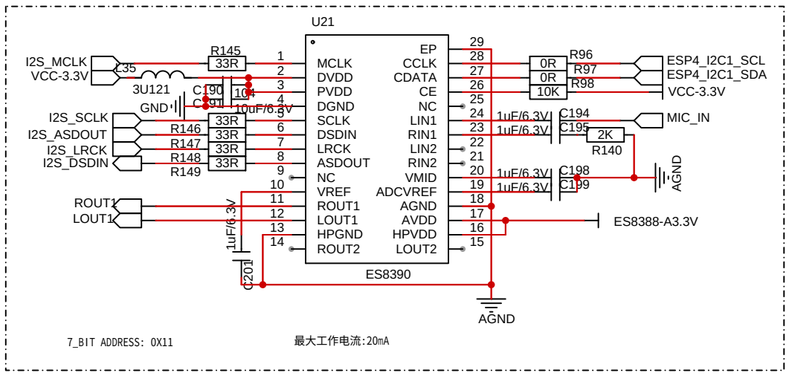
接收电流：5.7mA

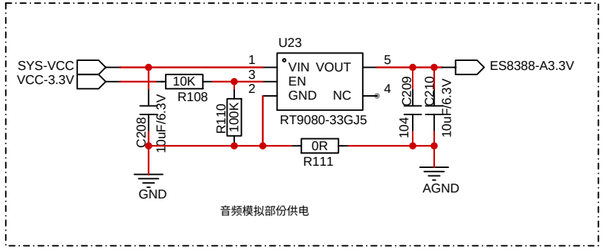
休眠电流：1.5uA

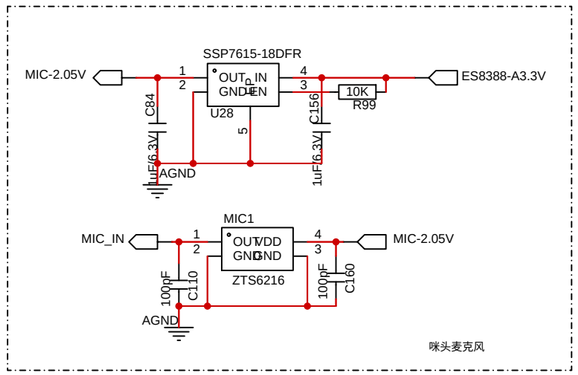

PJ-342
SPEAKER-OUT
R101
10R
C196
1uF/6.3V
ROUT1
C197
1uF/6.3V
LOUT1
R102
10R
PJ0
AGND
R103
470R
R104
470R
AGND
3.5耳机插座

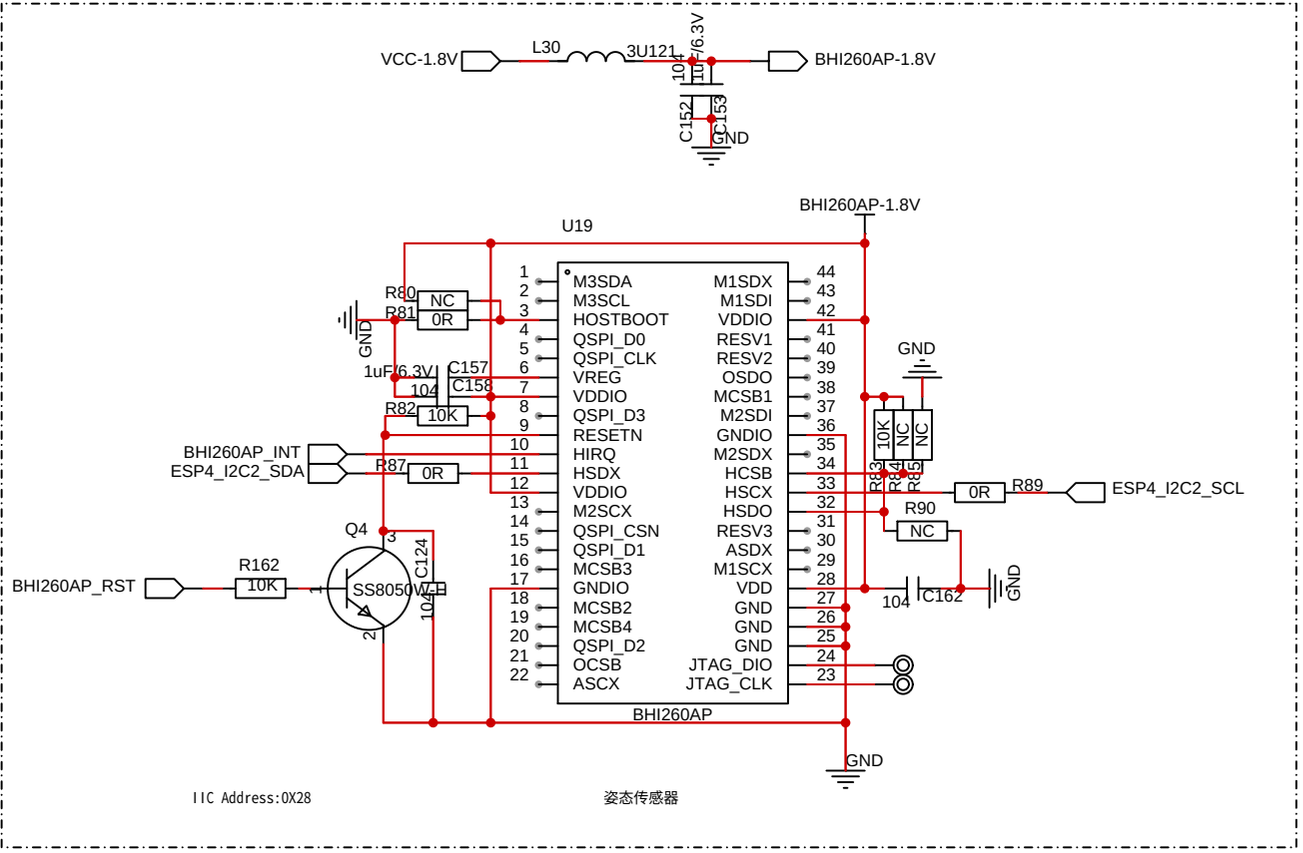

SPEAKER-CTL
SPEAKER-OUT
C203
1uF/6.3V
R106
30K
R107
30K
U22
CTRL
BYPASS
INP
INN
VOP
GND
VCC
VON
NS4150B
5D600
L36
NS-3.3V
L37
5D600
CN5
ZX-MX1.25-2PLT
GND
C204
1uF/6.3V
R109
10K
C205
1uF/6.3V
C206
100pF
C207
100pF
AGND
最大工作电流:350mA
8R喇叭功放


VCC-3.3V
L38
5D600
NS-3.3V
104
C211
10uF/6.3V
C212
C213
100uF/6.3V
AGND

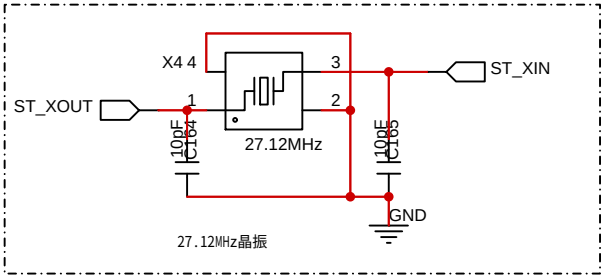

U21
I2S_MCLK
VCC-3.3V
C35
3U121
GND
R145
33R
C190
C191
104
10uF/6.3V
I2S_SCLK
I2S_ASDOUT
I2S_LRCK
I2S_DSDIN
R146
33R
R147
33R
R148
33R
R149
33R
MCLK
DVDD
PVDD
DGND
SCLK
DSDIN
LRCK
ASDOUT
NC
VREF
ROUT1
LOUT1
HPGND
ROUT2
EP
CCLK
CDATA
CE
NC
LIN1
RIN1
LIN2
RIN2
VMID
ADCVREF
AGND
AVDD
HPVDD
LOUT2
ES8390
R96
0R
R97
0R
R98
10K
ESP4_I2C1_SCL
ESP4_I2C1_SDA
VCC-3.3V
C194
1uF/6.3V
C195
1uF/6.3V
2K
R140
MIC_IN
AGND
C198
1uF/6.3V
C199
1uF/6.3V
ES8388-A3.3V
ROUT1
LOUT1
1uF/6.3V
C201
AGND
7_BIT ADDRESS: 0X11
最大工作电流:20mA

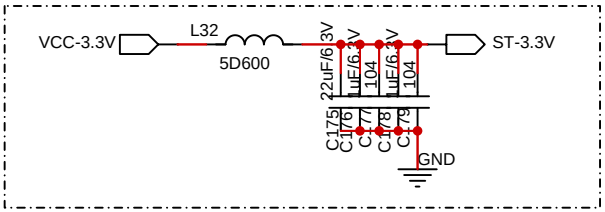

U23
SYS-VCC
VCC-3.3V
10K
R108
VIN
EN
GND
VOUT
NC
RT9080-33GJ5
ES8388-A3.3V
C208
10uF/6.3V
R110
100K
0R
R111
C209
104
C210
10uF/6.3V
GND
AGND
音频模拟部份供电

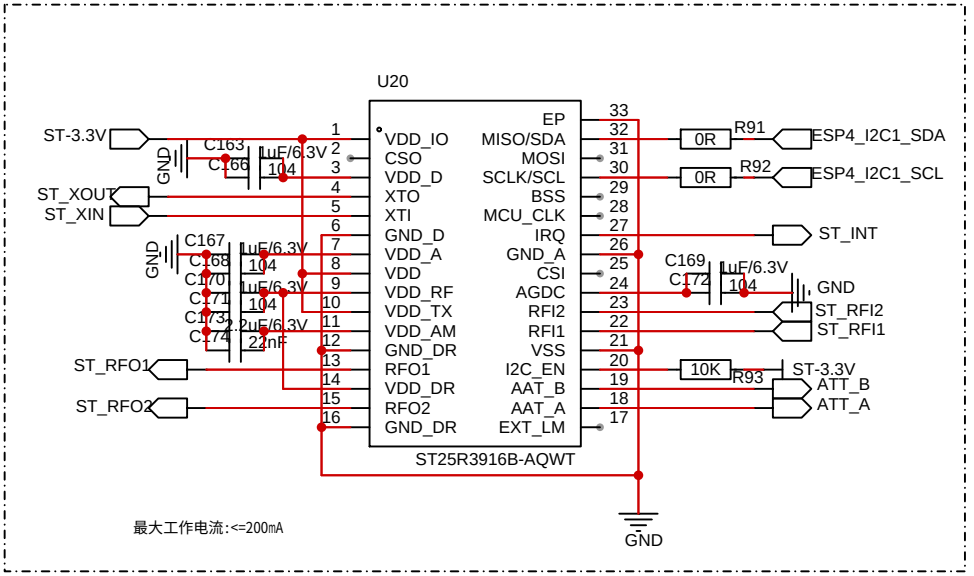

MIC-2.05V
SSP7615-18DFR
OUT
GND
IN
EN
U28
ES8388-A3.3V
10K
R99
C84
1uF/6.3V
AGND
C156
1uF/6.3V
MIC_IN
MIC1
OUT
GND
VDD
GND
ZTS6216
MIC-2.05V
C110
100pF
AGND
C160
100pF
咪头麦克风

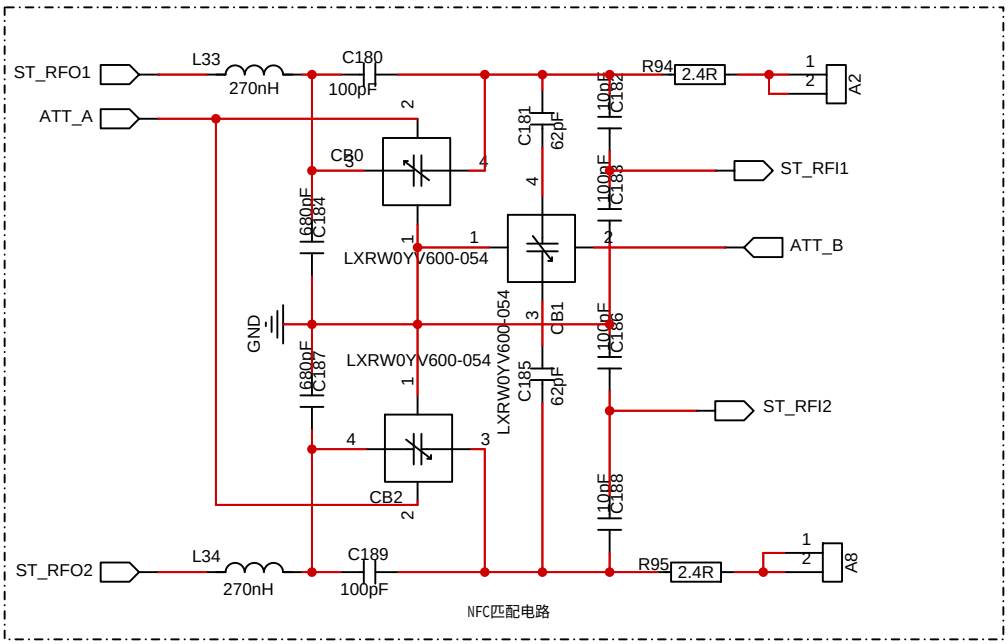

VCC-1.8V
L30
3U121
104
10uF/6.3V
BHI260AP-1.8V
C152
C153
GND
BHI260AP-1.8V
U19
R80
NC
R81
0R
GND
1uF/6.3V
C157
104
C158
R82
10K
BHI260AP_INT
ESP4_I2C2_SDA
R87
0R
Q4
R162
10K
BHI260AP_RST
SS8050W-H
C124
104
M3SDA
M3SCL
HOSTBOOT
QSPI_D0
QSPI_CLK
VREG
VDDIO
QSPI_D3
RESETN
HIRQ
HSDX
VDDIO
M2SCX
QSPI_CSN
QSPI_D1
MCSB3
GNDIO
MCSB2
MCSB4
QSPI_D2
OCSB
ASCX
M1SDX
M1SDI
VDDIO
RESV1
RESV2
OSDO
MCSB1
M2SDI
GNDIO
M2SDX
HCSB
HSCX
HSDO
RESV3
ASDX
M1SCX
VDD
GND
GND
GND
JTAG_DIO
JTAG_CLK
BHI260AP
GND
10K
NC
NC
R83
R84
R85
R89
0R
ESP4_I2C2_SCL
R90
NC
104
C162
GND
GND
IIC Address:0X28
姿态传感器
VCC-3.3V
L31
3U121
104
10uF/6.3V
QM-3.3V
C154
C155
GND
U18
GND
A1
0R
A2
R88
VSS SDA
SCL VDD
QMC6309
B2
R86
0R
B1
QM-3.3V
ESP4_I2C1_SCL
ESP4_I2C1_SDA
3D磁传感器

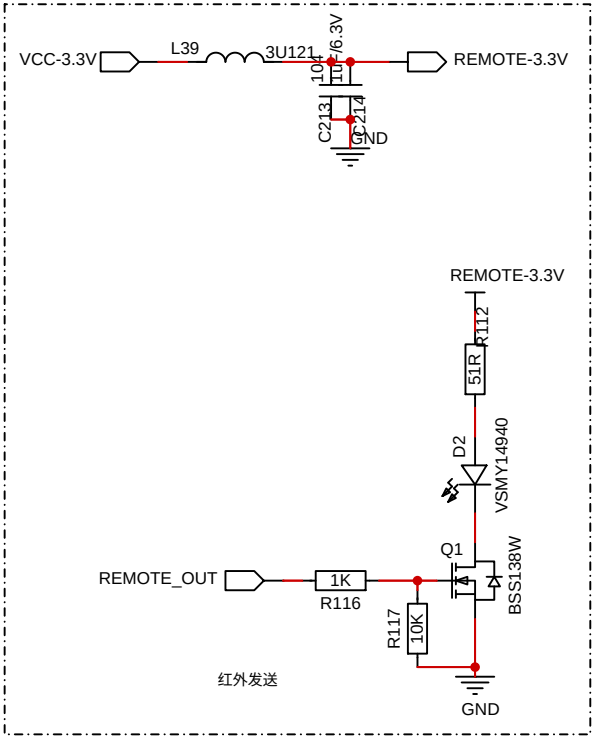
$$$14352 $$$22097

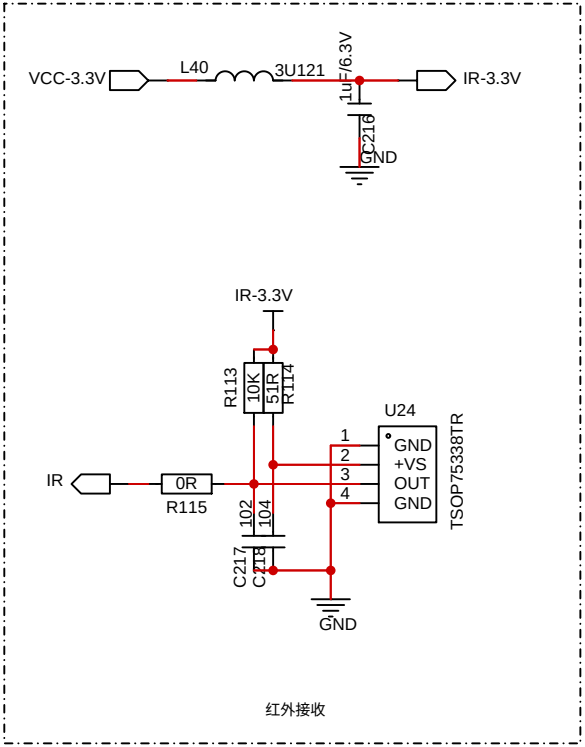

X4 4
ST_XOUT
ST_XIN
27.12MHz
C164
10pF
C165
10pF
GND
27.12MHz晶振

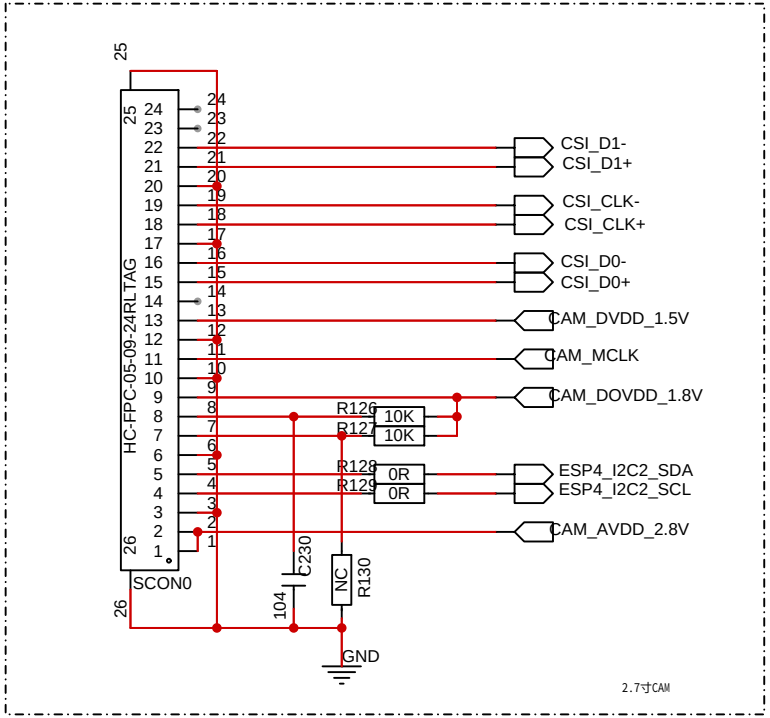

VCC-3.3V
L32
5D600
ST-3.3V
22uF/6.3V
1uF/6.3V
104
GND

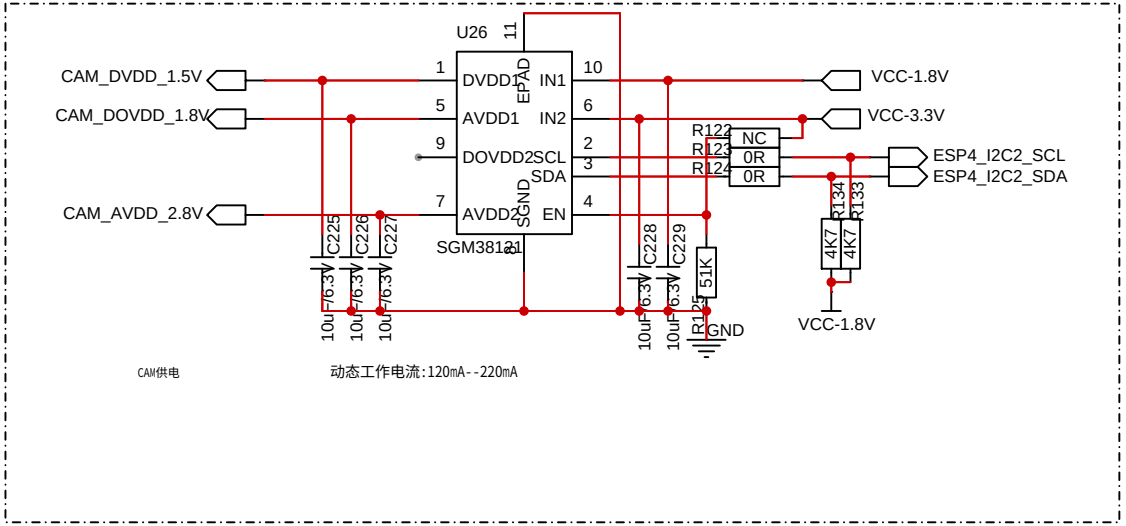

U20
ST-3.3V
ST_XOUT
ST_XIN
ST_RFO1
ST_RFO2
VDD_IO
CSO
VDD_D
XTO
XTI
GND_D
VDD_A
VDD
VDD_RF
VDD_TX
VDD_AM
GND_DR
RFO1
VDD_DR
RFO2
GND_DR
EP
MISO/SDA
MOSI
SCLK/SCL
BSS
MCU_CLK
IRQ
GND_A
CSI
AGDC
RFI2
RFI1
VSS
I2C_EN
AAT_B
AAT_A
EXT_LM
ST25R3916B-AQWT
R91
0R
ESP4_I2C1_SDA
R92
ESP4_I2C1_SCL
ST_INT
ST_RFI2
ST_RFI1
R93
10K
ATT_B
ATT_A
最大工作电流:<=200mA
GND

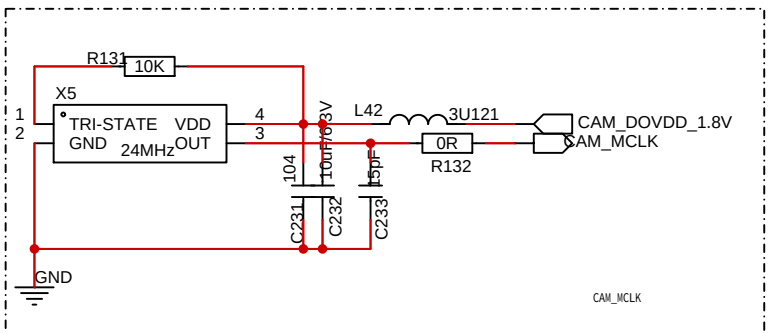

ST_RFO1
ATT_A
L33
270nH
C180
100pF
CB0
LXRW0YV600-054
CB1
LXRW0YV600-054
CB2
LXRW0YV600-054
C181
62pF
C185
62pF
C182
C183
C184
C186
C187
C188
R94
2.4R
A2
ST_RFI1
ATT_B
ST_RFI2
R95
2.4R
A8
ST_RFO2
L34
270nH
C189
100pF
GND
NFC匹配电路

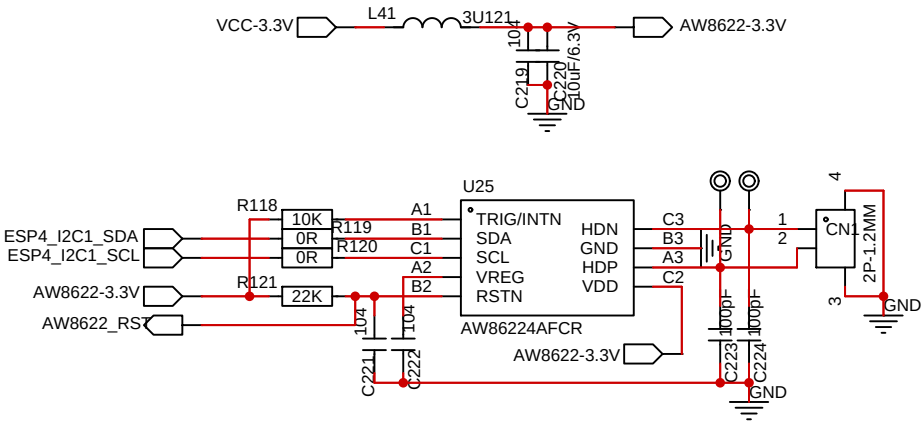

VCC-3.3V
L39
3U121
104
10uF/6.3V
REMOTE-3.3V
C213
C214
GND
REMOTE-3.3V
R112
51R
D2
VSMY14940
Q1
REMOTE_OUT
1K
R116
BSS138W
R117
10K
GND
红外发送


VCC-3.3V
L40
3U121
1uF/6.3V
IR-3.3V
C216
GND
IR-3.3V
R113
10K
51R
R114
U24
IR
0R
R115
1
2
3
4
GND
+VS
OUT
GND
TSOP75338TR
102
104
C217
C218
GND
红外接收


HC-FPC-05-09-24RLTAG
SCON0
CSI_D1-
CSI_D1+
CSI_CLK-
CSI_CLK+
CSI_D0-
CSI_D0+
CAM_DVDD_1.5V
CAM_MCLK
CAM_DOVDD_1.8V
CAM_AVDD_2.8V
ESP4_I2C2_SDA
ESP4_I2C2_SCL
R126 10K
R127 10K
R128 0R
R129 0R
C230 104
NC R130
GND
2.7寸CAM


U26
SGM38121
EPAD
SGND
DVDD1
AVDD1
DOVDD2
AVDD2
IN1
IN2
SCL
SDA
EN
CAM_DVDD_1.5V
CAM_DOVDD_1.8V
CAM_AVDD_2.8V
VCC-1.8V
VCC-3.3V
ESP4_I2C2_SCL
ESP4_I2C2_SDA
R122 NC
R123 0R
R124 0R
R134 4K7
R133 4K7
C225 10u/6.3V
C226 10u/6.3V
C227 10u/6.3V
C228 10uF/6.3V
C229 10uF/6.3V
R125 51K
GND
CAM供电
动态工作电流:120mA--220mA


R131 10K
X5
TRI-STATE
GND
VDD
24MHz
OUT
L42 3U121
0R R132
104 C231
10uF/6.3V C232
15pF C233
CAM_DOVDD_1.8V
CAM_MCLK
GND
CAM_MCLK



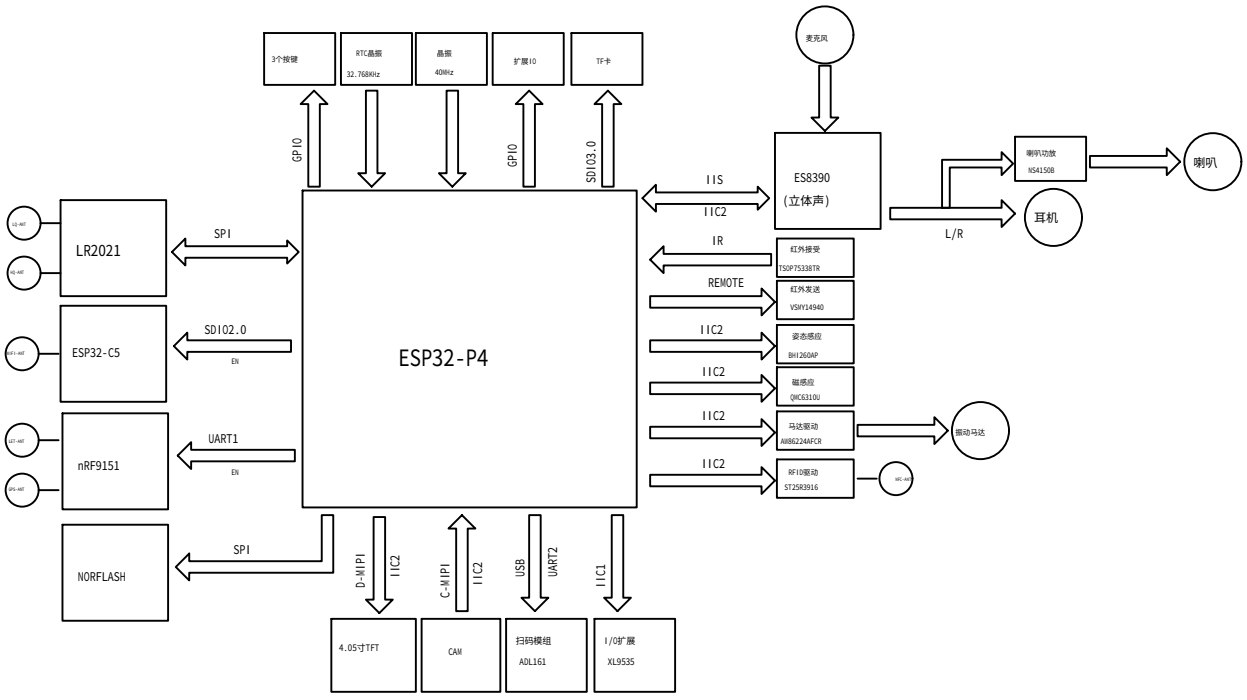

匹配马达额定工作电流:130mA

Standby工作电流:3uA

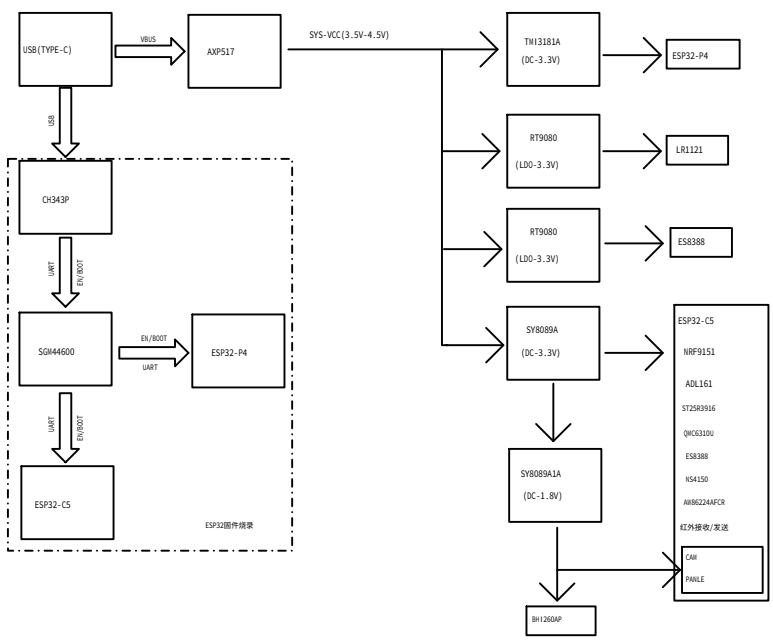
Shutdown工作电流:<1uA

Quiescent工作电流:1.5mA

马达驱动




BLOCK DIAGRAM




POWER DIAGRAM